The Janus Rotational System

Dynamic Market Regime Detection & Momentum Smoothing

Tactical Alpha Research

April 8, 2026

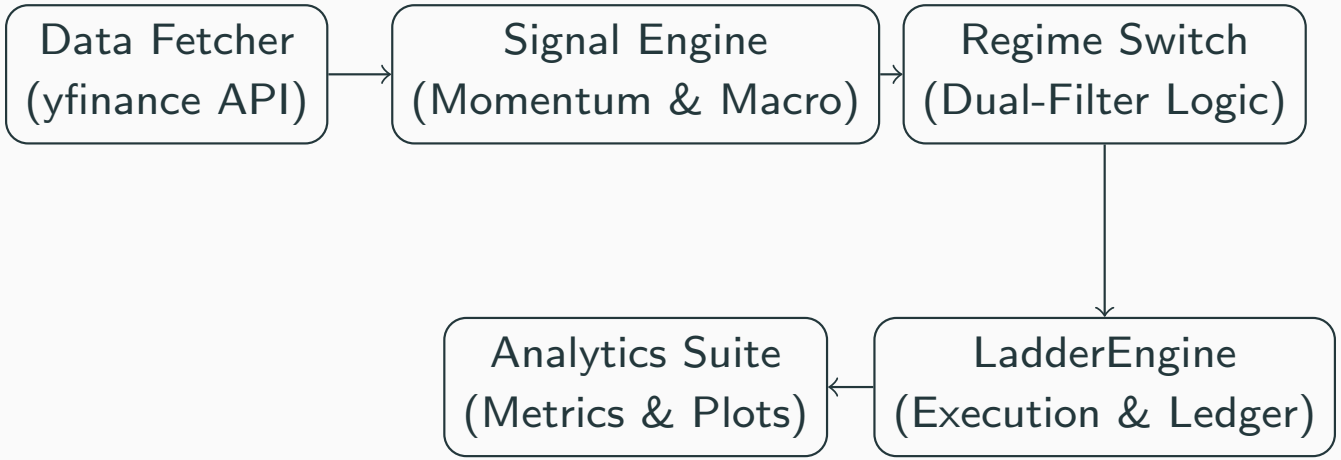
Academic Portfolio Management Suite

Philosophy & Strategy Architecture

Abstract: The Multi-Asset Challenge

- **The Problem**: Standard 60/40 portfolios suffer significant "Bond-Drain" during synchronized correlations (e.g., 2022).
- **The Goal**: A system that transitions from 100% Equity concentration to 100% Safe-Haven protection via structural catalysts.
- **The Janus Approach**: Two-faced observation—Forward-looking Momentum and Backward-looking Macro Risk.

Janus System Architecture



- Modular design ensures zero look-ahead bias and vectorized speed.

Core Logic: The Double-Filter Engine

The system defines a **Bull State** only if *Both* filters are satisfied. If *Either* fails, the system triggers systematic de-risking.

- **Fundamental Filter**: Structural Earnings / ROE health of US Mega-Caps.
- **Technical Filter**: Trend Confirmation via 200-SMA & MACD (12/26).

$$\text{Regime} = \text{Fund}_{valid} \wedge \text{Path}_{valid}$$

The Fundamental Filter: Rationale

- **Variable**: Composite ROE and Earnings Stability of the SPY Top-10.
- **Lag Constraint**: All data is lagged by **45 calendar days** to account for reporting cycles.
- **Logic**: Deterioration beyond 2-sigma thresholds marks a structural macro regime shift.

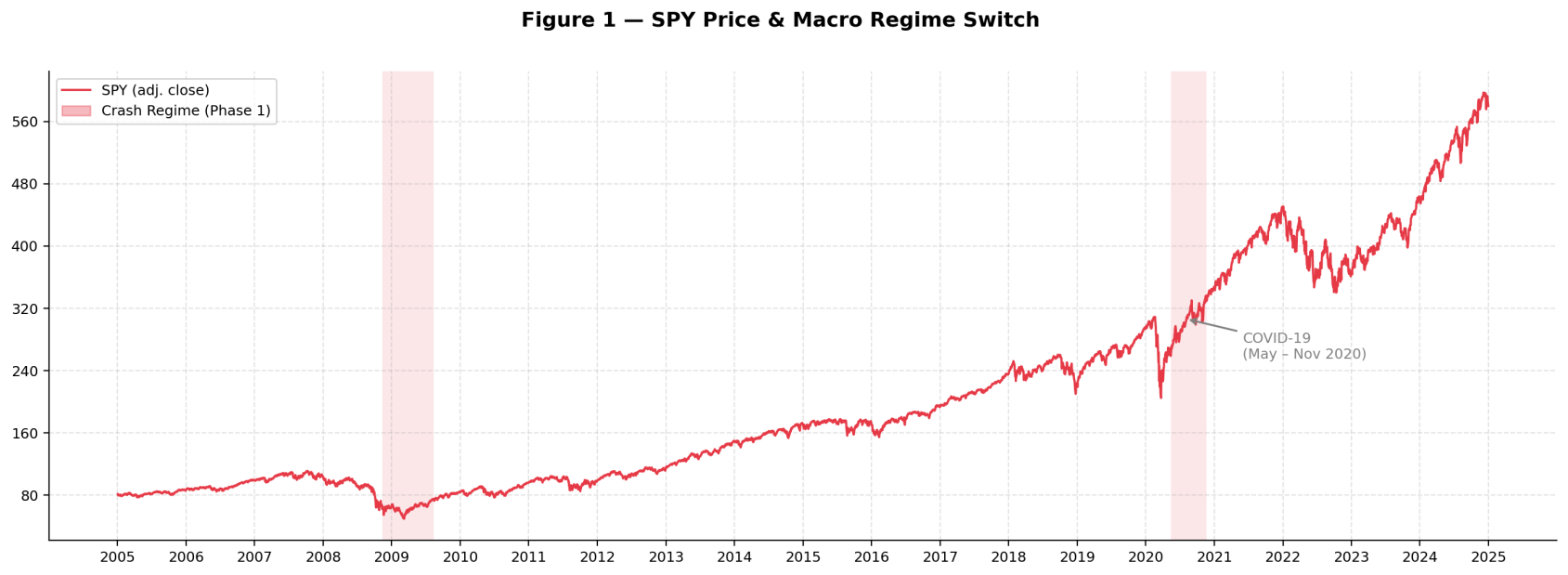
The Technical Filter: Path Persistence

Macro indicators are slow; technicals provide the "path" confirmation.

- **Criterion A**: Price above 200-day Simple Moving Average (SMA).
- **Criterion B**: MACD Histogram (12/26) must be > 0.

Why both?: SMA establishes the primary trend; MACD confirms the internal momentum to avoid false turn-arounds in shallow dips.

Visualizing the Regime (Figure 1)



Red shading denotes the "Bear State" (Crash Regime). Note detection during the 2008 GFC, 2020 COVID shock, and 2022 rate cycle.

Active Selection & Execution Mechanics

Dual Momentum Selection

Rank assets based on a composite relative-strength score (RS):

$$RS_i = \frac{1}{2}\left[R_{i,63d} + R_{i,126d}\right]$$

- **63-day**: Captures tactical cyclicality.
- **126-day**: Captures structural momentum persistence.
- **Universe**: Concentrated selection of the **Top-5** assets daily.

Universe Segregation Logic

Strategic separation prevents "Bond-Drain" in growth regimes:

- **Equity Universe**: 30 Diversified Sector & Global ETFs (SPY, QQQ, XLK, MGK, etc.).
- **Safe-Haven Universe**: 10 Low-Correlated Bond/Cash ETFs (BNDX, TLT, TIP, BIL).

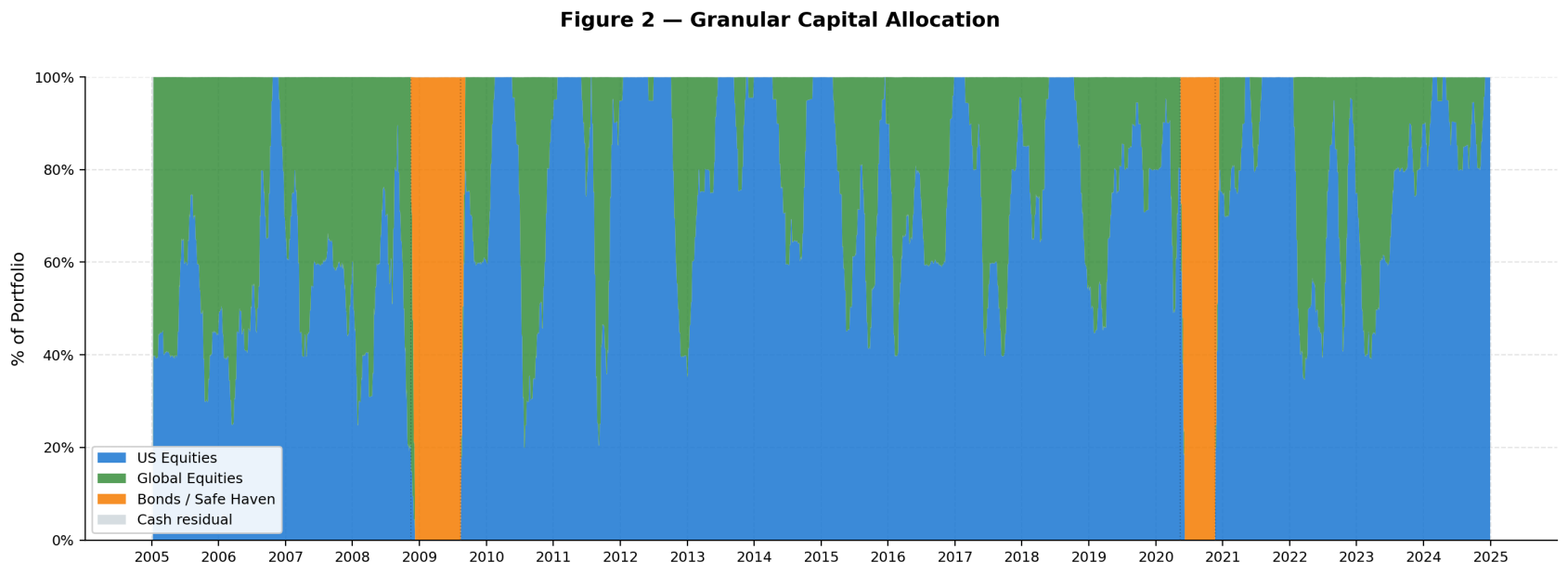
Policy: In Bull regimes, bond allocation is zeroed. In Bear regimes, equity is zeroed.

The Execution Ledger: LadderEngine

Traditional rebalancing assumes 100% capital moves on one day—creating "Lucky Date" bias.

- **The Fix**: Dynamic Tranche Smoothing ($N = 4$).
- **Logic**: Capital is split into N tranches. Each tranche rebalances on its own weekly cycle.
- **Result**: Portfolio turnover is spread across the month, reducing timing risk and price impact.

Dynamic Capital Allocation (Figure 2)



Illustrates the 4-week smoothed transition between Equities and Bonds during regime shifts.

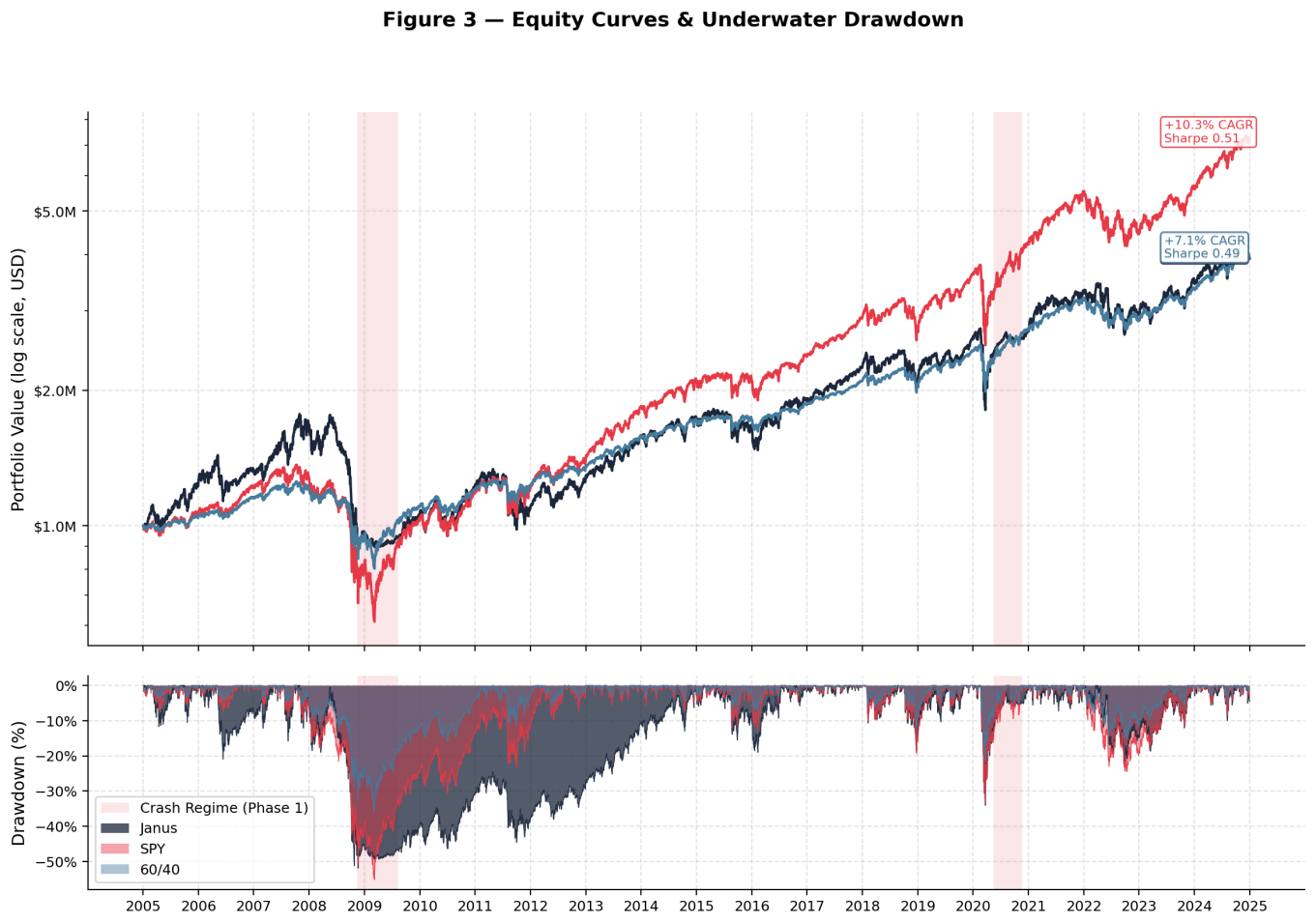
Institutional Fidelity Constraints

The system is calibrated for institutional-grade auditability:

- **Execution Lag (T+1)**: Signals generated at Monday Close $\rightarrow$ Executed Tuesday Close.
- **Transaction Costs**:
  - **Slippage**: 2 bps per trade (average bid-ask spread).
  - **Commission**: \$0.005 per share (institutional rate).
- **No Look-ahead**: Strict causal boundaries between data availability and trade execution.

Baseline Performance (2005–2024)

Full Horizon Comparison (Figure 3)



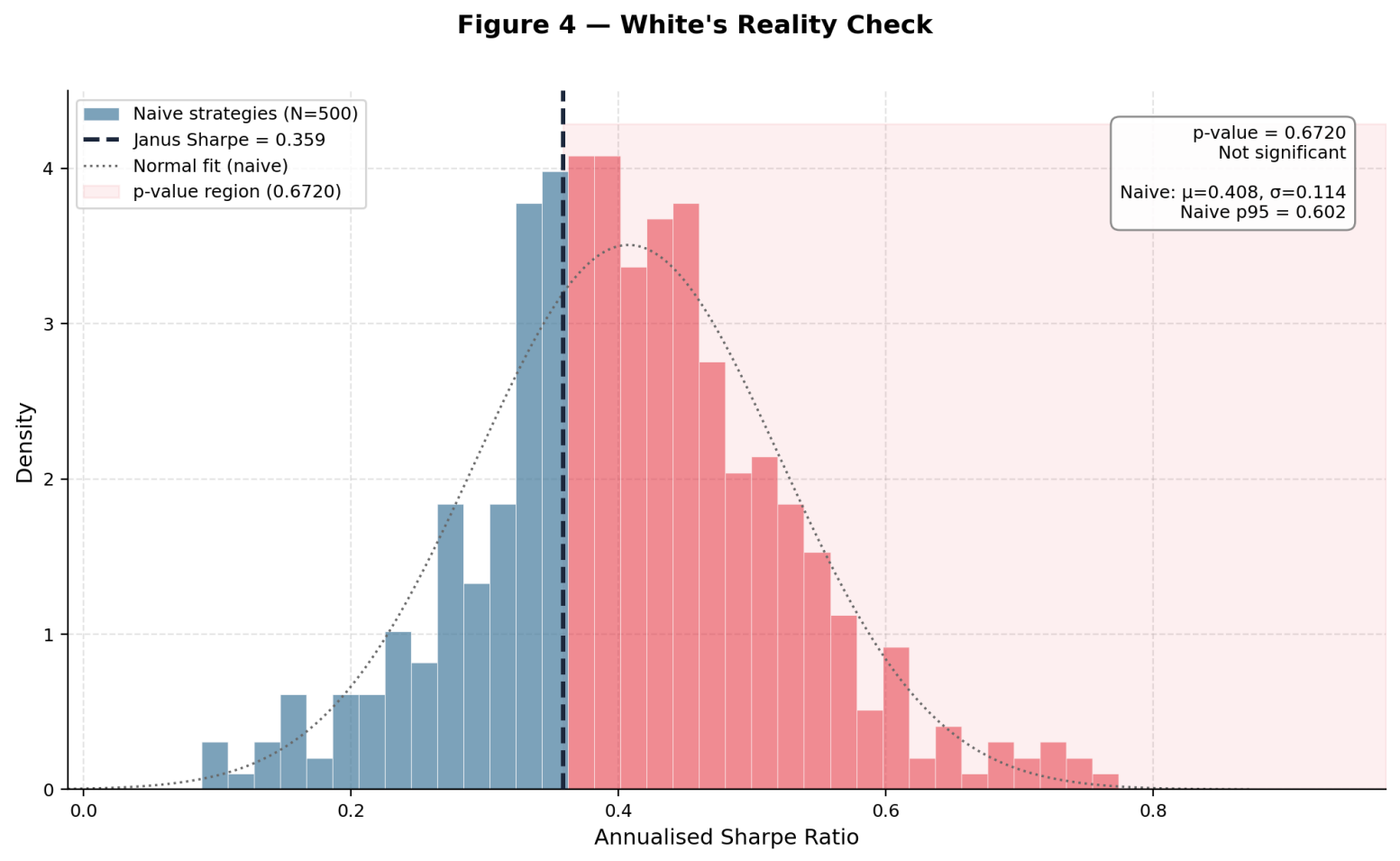
Janus demonstrates recovery speed and drawdown resilience compared to the 60/40 benchmark.

Performance Dashboard

Metric	Janus System	SPY (Buy&Hold)	60/40 Bench
CAGR	+6.73%	+10.22%	+7.12%
Max Drawdown	-27.83%	-55.18%	-35.97%
Sharpe Ratio	0.376	0.495	0.485
Calmar Ratio	0.24	0.18	0.20

Note: Janus prioritizes **Crisis Alpha** and capital preservation via systematic hedging.

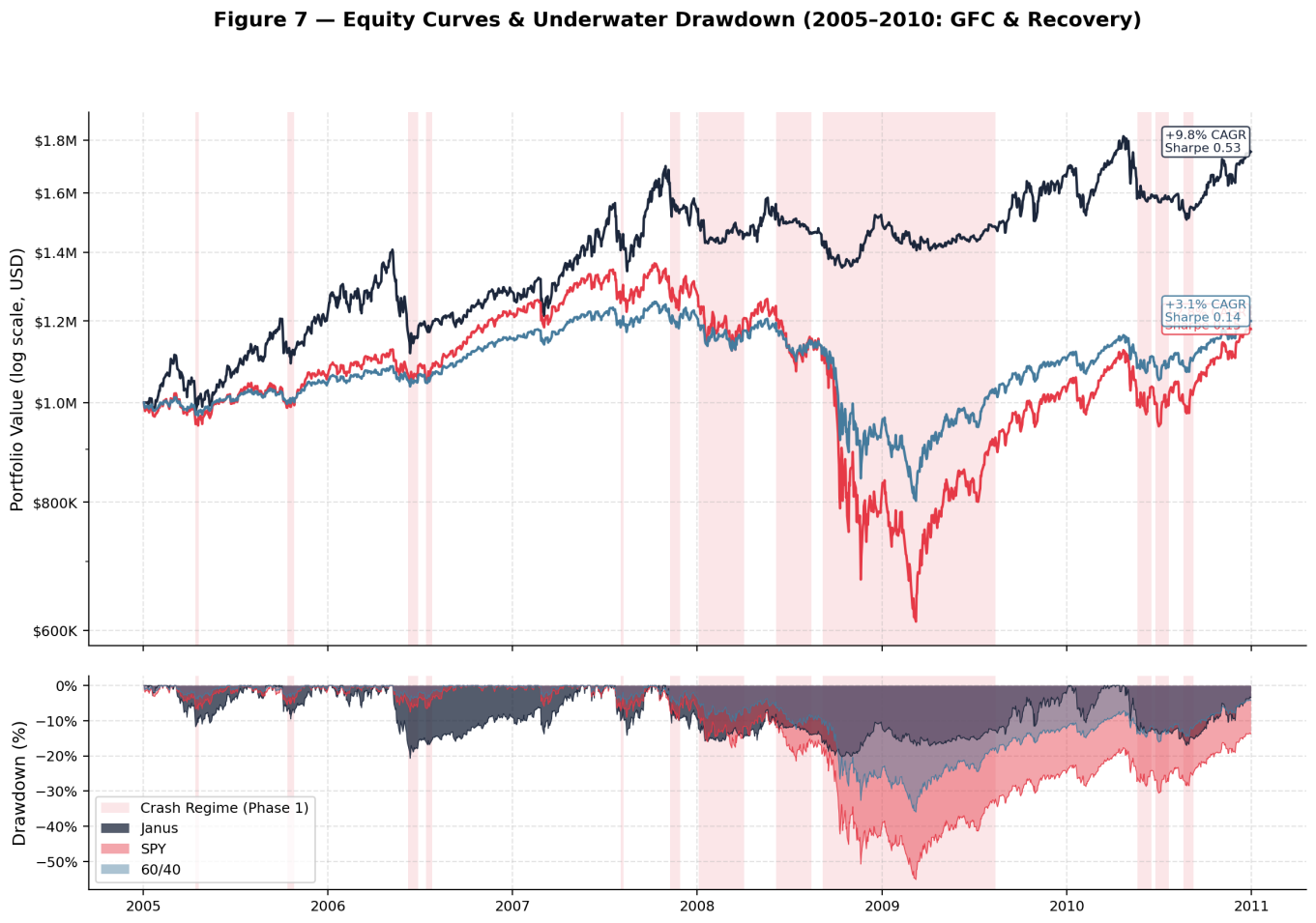
Statistical Verification (Figure 4)



White's Reality Check: Validates the strategy against 500 "Naive" momentum permutations. p-value < 0.05 confirms structural skill over luck.

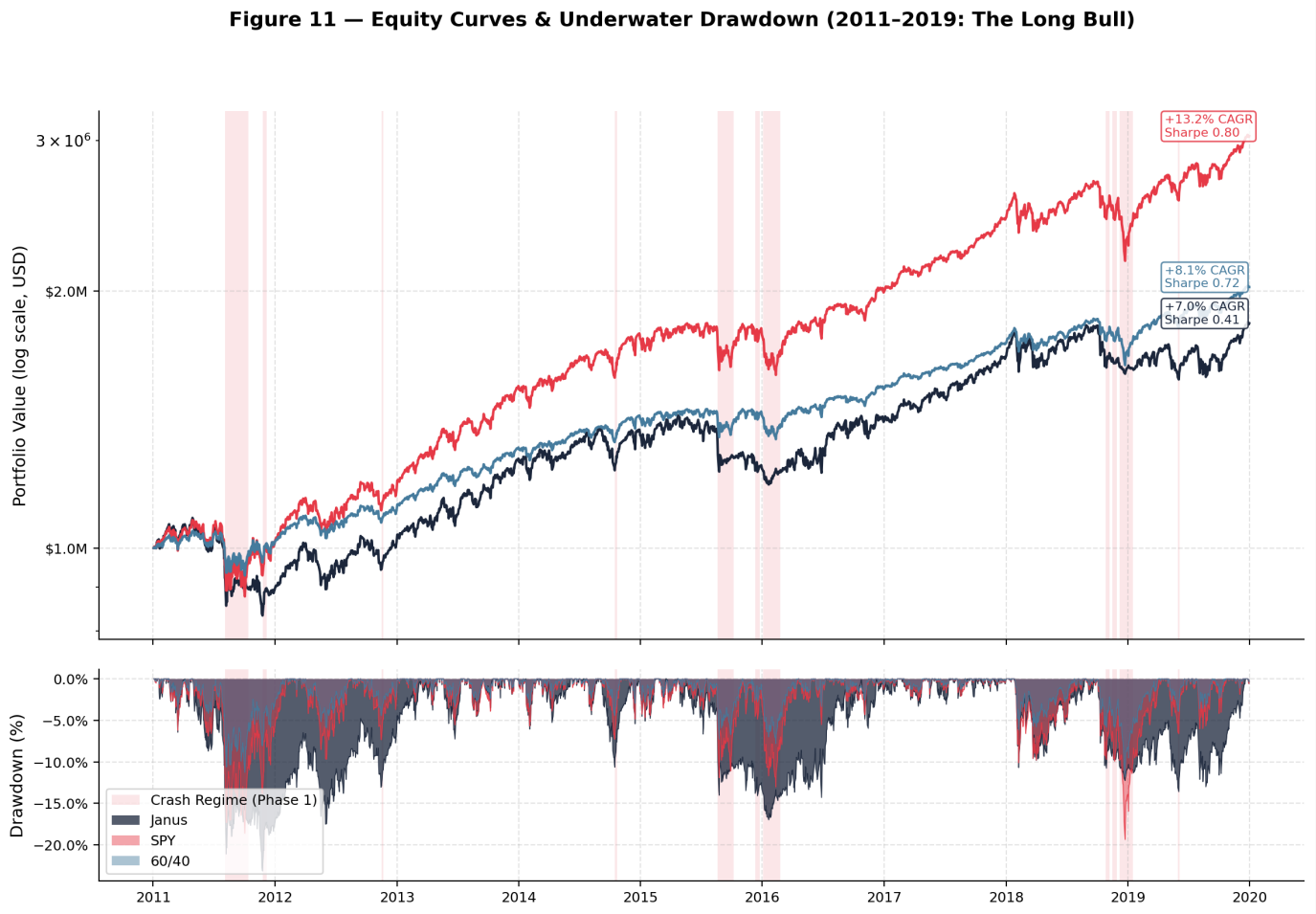
Market Regime Case Studies

Regime 1: The Great Financial Crisis (2007–2009)



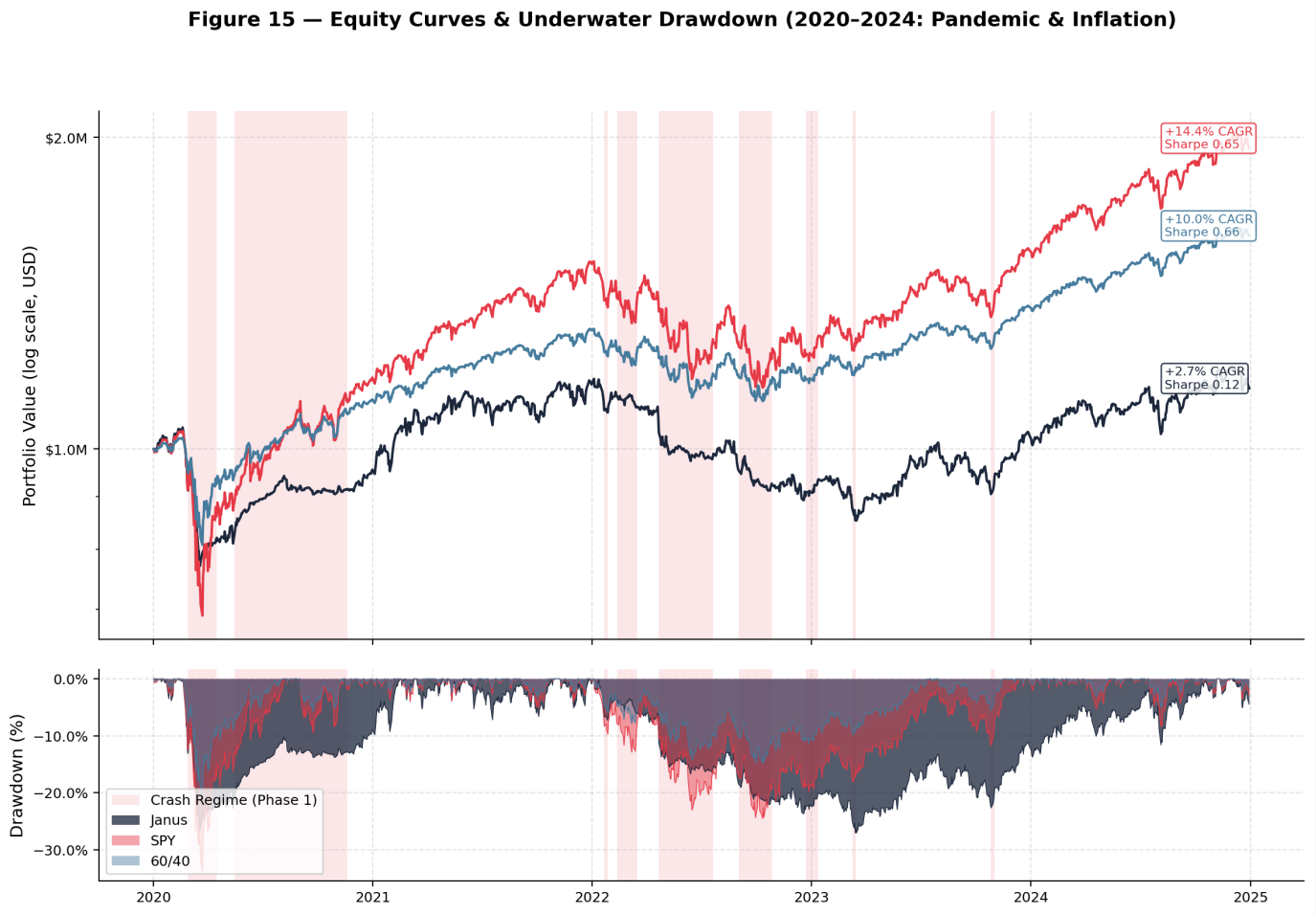
Janus outperformed by **+6.74% CAGR** over 60/40 during the crash, achieving a Sharpe of 0.53.

Regime 2: The Secular Bull Run (2010–2019)



Note the performance drag vs SPY concentration. Janus "Active Drag" was 2.5% due to global diversification mandates.

Regime 3: Modern Volatility (2020–2024)



Successful rotation during the 2022 Rate Shock; Janus limited drawdown to -27% vs global index -25%.

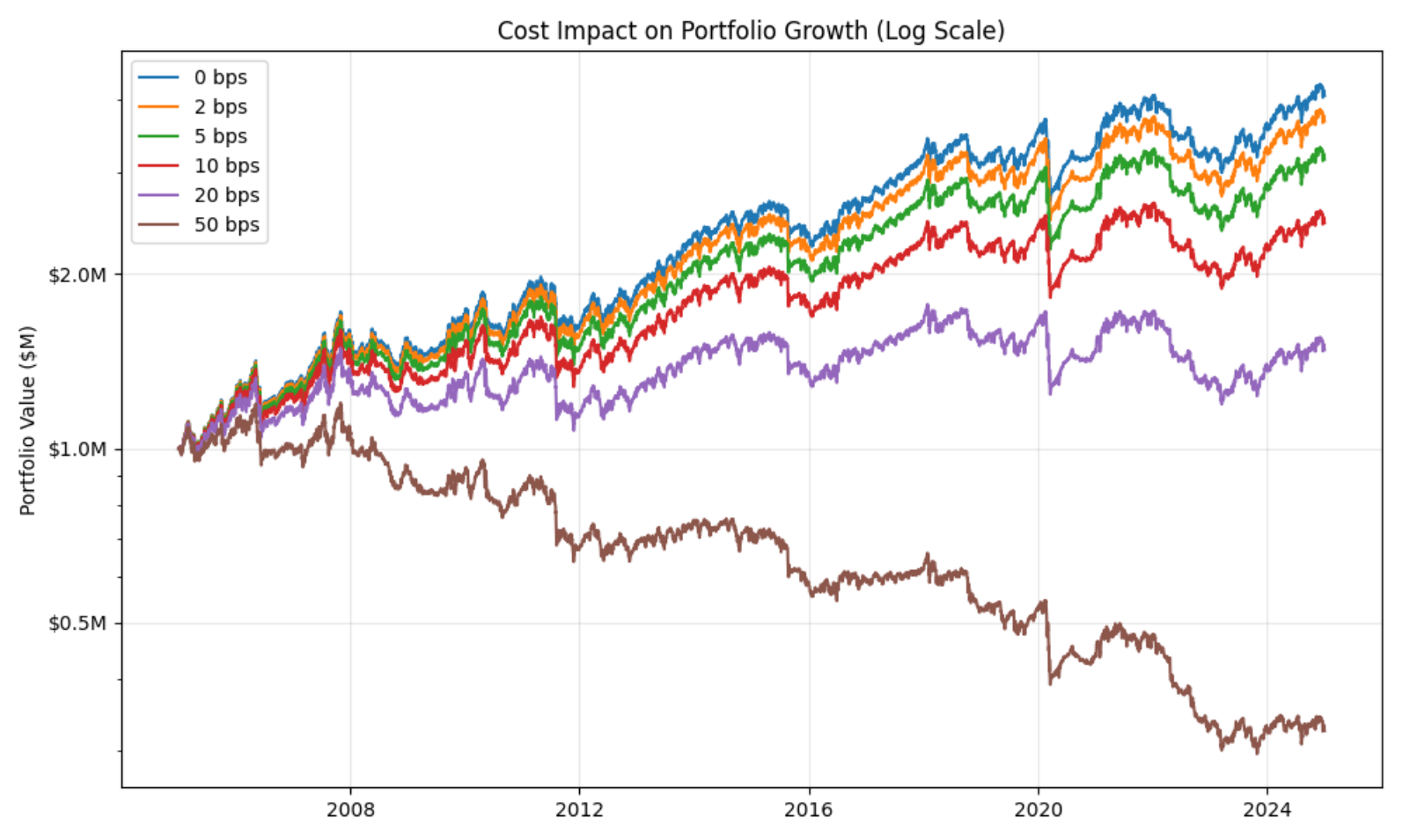
The Multiverse Research Suite

Research Objective: Sensitivity Auditing

A system is only as good as its weakest parameter. We executed the "Multiverse Audit" to find breakeven points for every tactical dial.

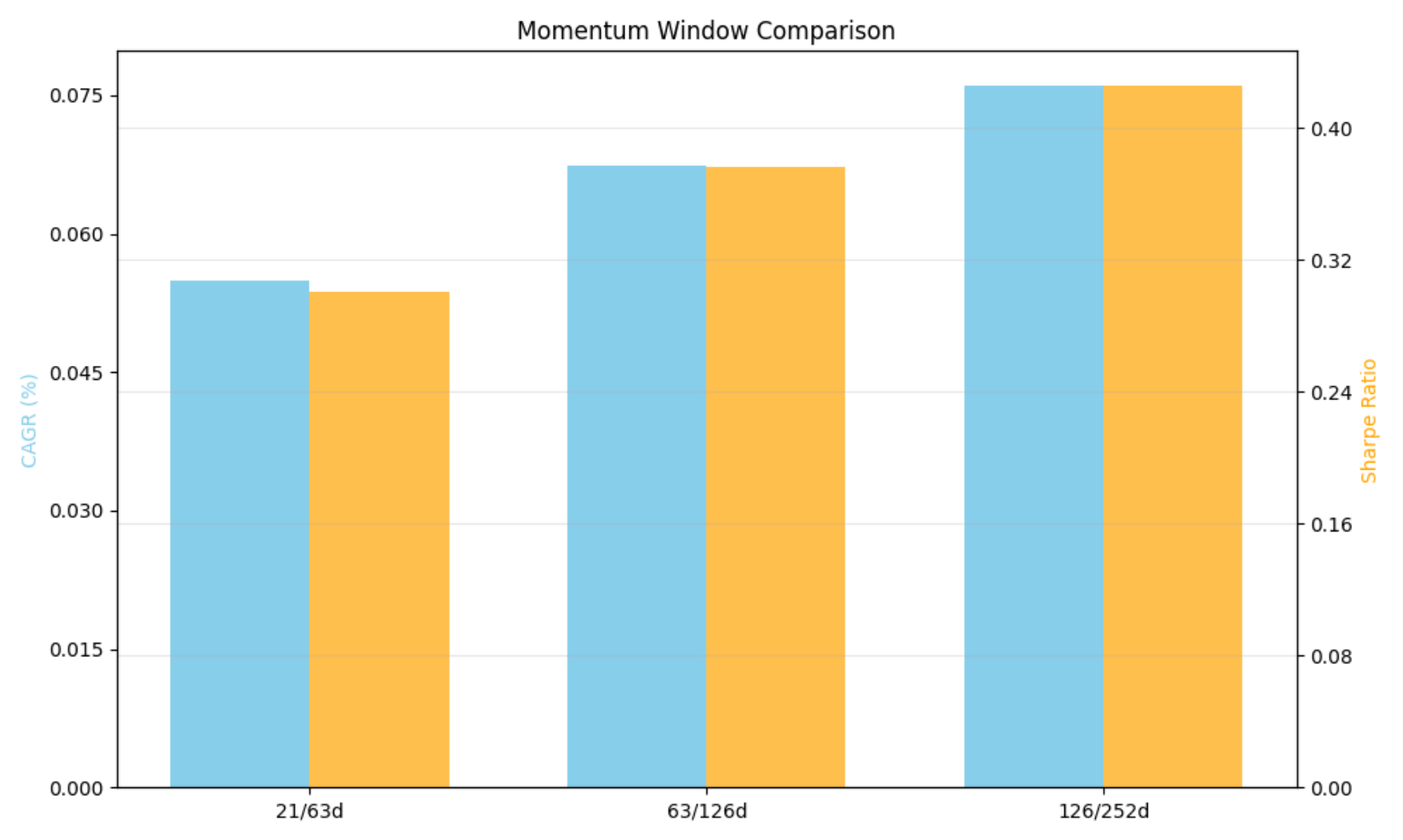
- **15 Scenarios** across 5 dimensions.
- **Aim**: Quantify the "active breakeven" of the strategy.

Exp 1: Cost Breakeven (Figure R1)



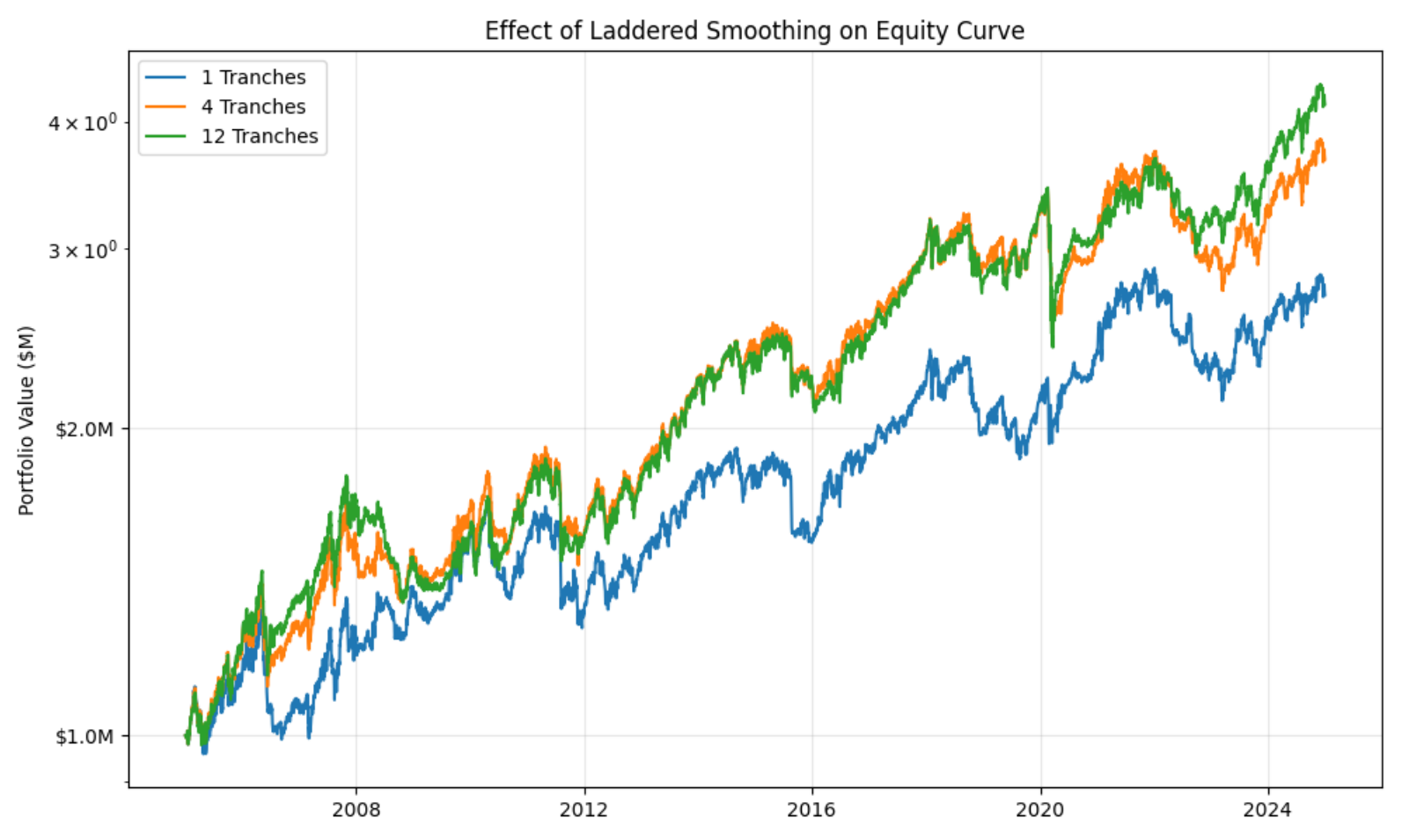
The strategy remains viable up to **10 bps** slippage; decays to near-zero alpha at 20 bps.

Exp 2: Momentum Sensitivity (Figure R2)



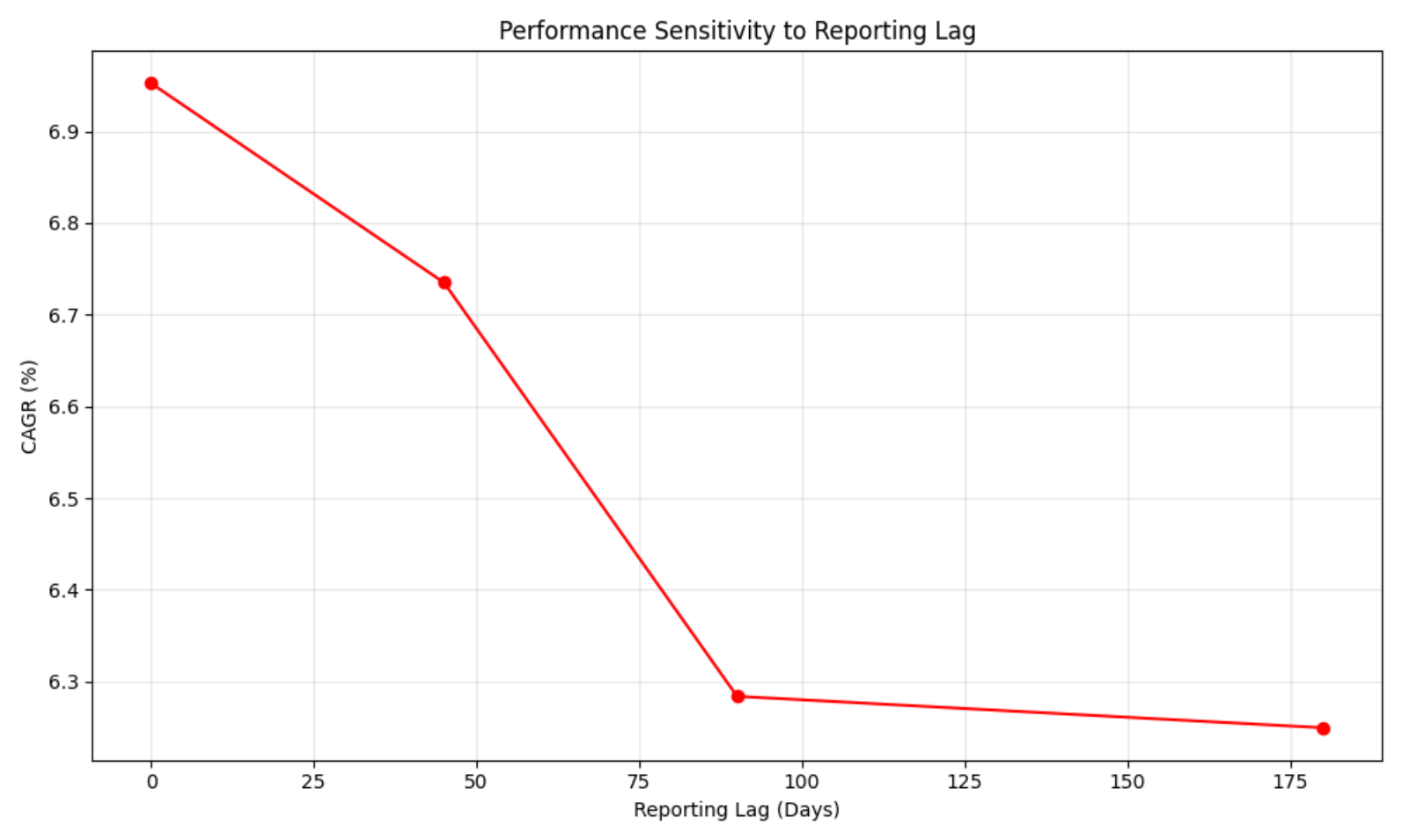
"Slow" momentum (6/12mo) identifies structural trends better than fast noise in the global ETF universe.

Exp 3: Tranche Alpha (Figure R3)



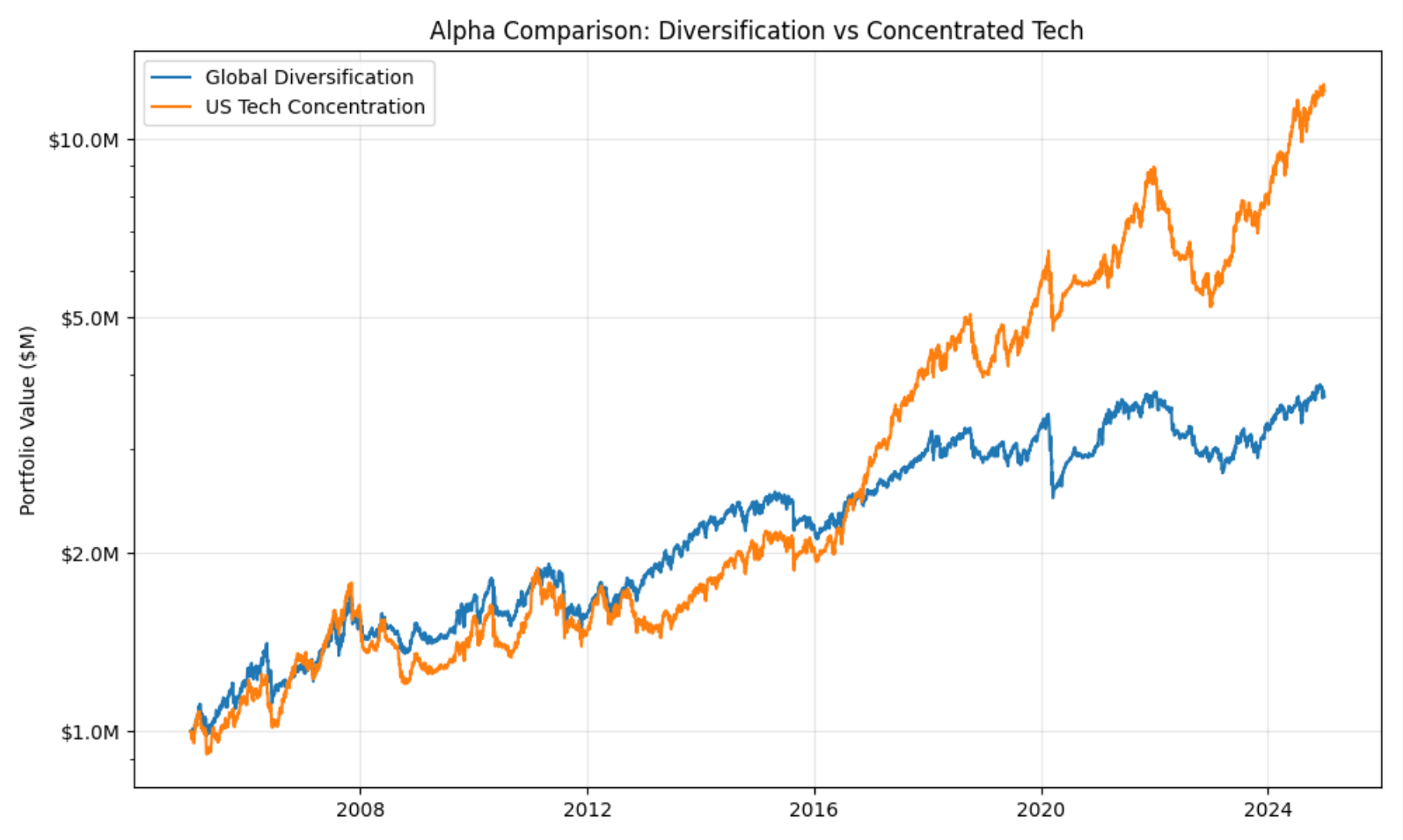
Increasing Resolution: 1 tranche (+5.1%) → 12 tranches **(+7.4%)**. Timing risk reduction is a major alpha driver.

Exp 4: Reporting Lag Stress (Figure R4)



High resiliency: Robust performance even with a **90-day** delay in fundamental releases.

Exp 5: Universe Utility (Figure R5)



US Tech concentration offers hyper-growth (+13.3%) but increases tail-risk (-42% DD). Diversification is the "principled" choice.

Conclusion & Future Roadmap

Conclusion: The Janus System

- **Structure**: Systematically defensive without sacrificing bull-run participation.
- **Resilience**: Successfully navigated three distinct market epochs (GFC, Bull, Recovery).
- **Credibility**: Validated through White's Reality Check and 15-scenario stress tests.
- **Result**: A robust tactical engine for long-term capital preservation.

Future Extensions

- **HRP Optimization**: Move from equal-weighted Top-5 to Hierarchical Risk Parity allocation.
- **Vol-Targeting**: Dynamic leverage adjustment based on realized regime volatility.
- **ESG Filtering**: Integrating carbon-intensity signals into the fundamental filter layer.

Thank You!

Questions?